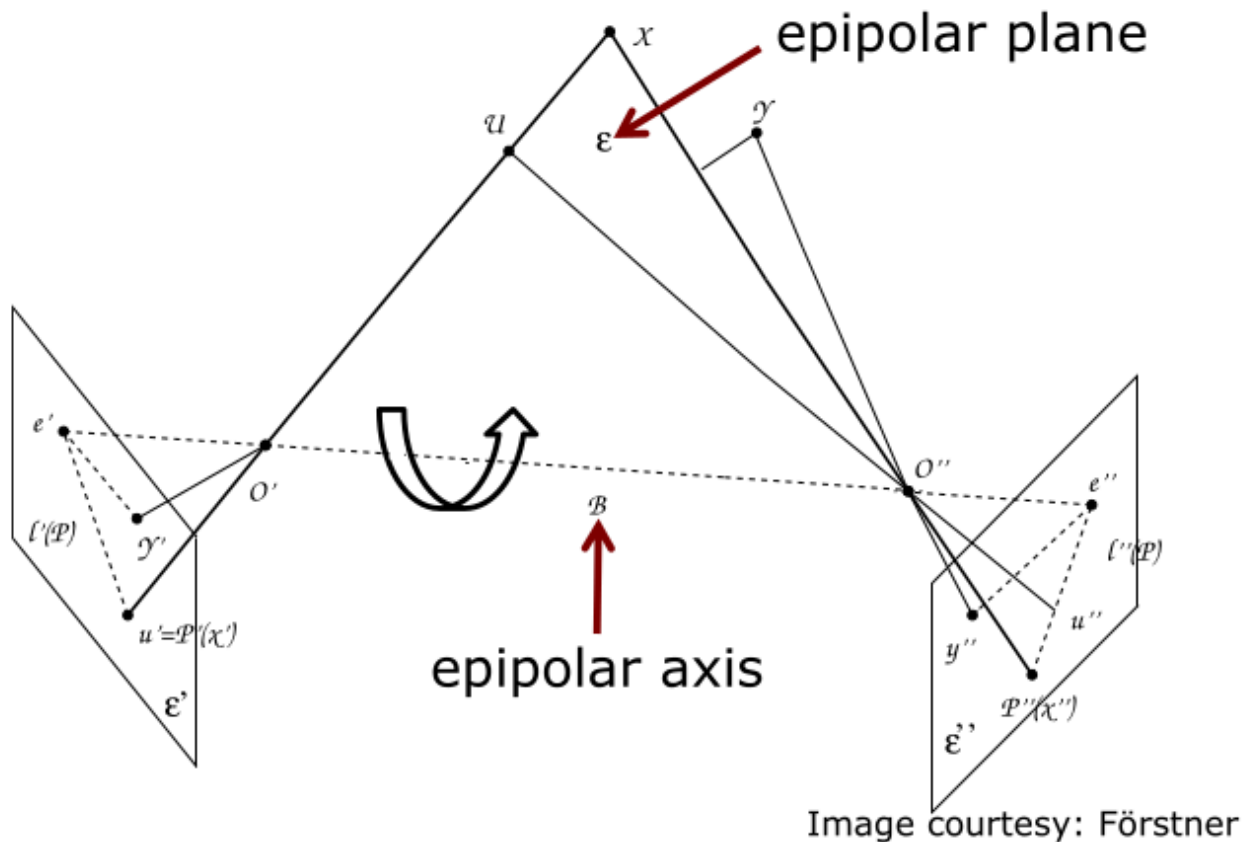


L08: Epipolar Geometry

- Describes geometric relations in image pairs
- Enables efficient search for corresponding points
- Given straight-line preserving mapping, search space reduces from 2D(whole image) to 1D (line)
- **Given** point $\mathcal{X}$ in the plane of the **first** image
- **Find** the corresponding point $\mathcal{X}''$ in the **second** image plane



Important Elements

- **Epipolar axis:** $B = (O'O'')$
 - line through the two projection centers
- **Epipolar plane:** $\epsilon = (O'O''X)$
 - depends on the projection centers and the point
- **Epipoles:** $e' = (O'')$, $e'' = (O')$
 - images of the projection centers
 - in the other image respectively
 - alternatively written:
 - $e' = (O'O'') \cap E'$ and $e'' = (O'O'') \cap E''$

- **Epipolar lines:** $l'(X) = (O''X)', l''(X) = (O'X)''$
 - images of rays $(O''X)$ and $(O'X)$
 - in the other image respectively
 - alternatively written:
 - $l'(X) = \epsilon \cap \epsilon'$ and $l''(X) = \epsilon \cap \epsilon''$

Epipolar Plane

- Using **distortion-free lens**
- All lie in the epipolar plane ϵ
 - the projection centers O', O''
 - the point X
 - the epipolar lines $l'(X), l''(X)$
 - the epipoles e', e''
 - the image points X', X''
- simplifies predicting the location of the corresponding point in the other image

Find the location of the corresponding points

- Task: Predict location X'' given X'
- The epipolar plane through $\epsilon = (O'O''X')$
- Intersection of epipolar plane and second image plane ϵ'' yields the epipolar line $l''(X)$
- Corresponding point X'' lies on the epipolar line $l''(X)$

Reduces the search space from 2D to 1D

Computing the key Elements of the epipolar Geometry

- **Epipolar Axis:** From projection centers
 - Homogenous vector $b = X_{O''} - X_{O'}$
 - **only direction, not length**
- **Epipolar Lines:** Image points on epipolar lines
 - Exploit coplanarity constraint for both points
 - $x'^T \frac{Fx''}{l'} = 0$
 - $x'^T \frac{Fx''}{l''} = 0$

◦  $l' = Fx''$ $l'' = F^T x'$ **Prediction (1D space)**

- **Epipoles:** Projection Centers in other image

- From projection Matrices

$$\blacksquare e' = P' X_{O''} \quad e'' = P'' X_{O'}$$

- Is the Intersection of ALL Epipolar Lines in an Image

$$\blacksquare \forall l' : e'^T l' = 0 \quad \forall l'' : e''^T l'' = 0$$

- $e'^T F x'' = 0$

- Epipoles is null space of F^T

- $x''^T F'^T e' = 0$

Eigenvectors of F

- Epipoles are left and right eigenvectors of the fundamental matrix

$$\begin{aligned} e'^T F &= 0 & F e'' &= 0 \\ \text{null}(F^T) &= e' & \text{null}(F) &= e'' \end{aligned}$$

Epipolar Geometry **reduces search for correspondences** in image pairs from 2D space to 1D space